



Shortest Path, Homework 12

1 Shortest-Paths Properties.

1. Let G be a directed graph, $\delta(u, v)$ the length of a shortest path from u to v , and $d[v]$ an estimator for $\delta(s, v)$ for a fixed source $s \in V$. Assume that $d[v]$'s are initialized as follows:

$$d[v] = \begin{cases} \infty & v \neq s \\ 0 & v = s \end{cases}$$

And that $d[v]$ values are updated by relaxations only. Prove on your own the **Upper-bound property** stated as follows: After the application of any sequence of relaxation steps, we have $d[v] \geq \delta(s, v)$.

2 Bellman-Ford

1. Let G be a directed, connected graph that may contains negative cycles. Prove that if G is indeed contains such then Bellman-Ford detects it. An implementation of Bellman-Ford is given in Algorithm 1.

Algorithm 1: Bellman–Ford.

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1 for  $v \in V$  do
2    $d[v] \leftarrow \infty$ 
3    $\pi[v] \leftarrow \text{null}$ 
4 end
5  $d[s] \leftarrow 0$ 
6 for  $i \in [n - 1]$  do
7   for  $(u, v) \in E$  do
8     if  $d[v] > d[u] + w(u, v)$  then
9        $d[v] \leftarrow d[u] + w(u, v)$ 
10       $\pi[v] \leftarrow u$ 
11    end
12  end
13 end
14 for  $(u, v) \in E$  do
15   if  $d[v] > d[u] + w(u, v)$  then
16     return Negative Cycle
17   end
18 end

```
